Pramesh Sharma

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Skills

- Python
- C++
- TensorFlow
- Google Cloud
- Linux
- JavaScript
- Go
- PyTorch
- Docker
- Git

Awards

Regents Scholar

AvenueE Scholar

Engineering, Math, and Physics Scholar

Michael Sager Memorial Scholar

Deloitte Leadership, Allyship & Mentorship Program

Genentech gLINX Program

St. Jude BioHackathon Travel Award

Posters

A Method for Detecting Conformational Changes in Fold-Switching Proteins
National Institutes of Health Postbac Poster Day
Pramesh Sharma, Vitaliy Kurlin, Lauren L. Porter

Education

University of California, Davis — Davis, CA
Bachelor of Science (Computer Science) — 09/2021 – 06/2024

Experience

National Institutes of Health — Bethesda, MD
Postbaccalaureate Fellow — 08/2025 – Present

- Developed an AI interpretability tool leveraging AlphaFold attention weights via JAX to extract biologically meaningful insights from protein residue mutations resulting in co-authorship on a bioRxiv preprint
- Engineered an inference pipeline integrating distance-based unsupervised clustering on backbone geometry atoms to identify fold-switching proteins across subsampled multiple sequence alignment predictions
- Streamlined automated testing, multi-environment package releases, and container builds across different Python versions with GitHub Actions workflows reducing deployment time by 3 minutes per pull request
- Instituted an elevated standard for code quality increasing test coverage to 90% using pytest and codecov for coverage reports on a codebase spanning 70+ unit, integration, and end-to-end tests

Intelligent Retail Lab by Walmart — Sunnyvale, CA
Software Engineer — 07/2024 – 07/2025

- Optimized collection of hardware metrics across cluster of NVIDIA Jetson GPUs by replacing jetson-stats through development of a real-time system monitoring tool and cloud dashboard using Python, Prometheus, and Grafana
- Automated machine learning training workflows by developing a model converter that transforms PyTorch ONNX models into DeepStream-compatible TensorRT engines across store vision systems using Kubeflow and WandB
- Improved image transformation processing for 40,000+ images of grocery items by building a scalable data augmentation pipeline in Python using Apache Beam and Dataflow deployed to Artifact Registry as Google Flex Template
- Reduced object detection inference latency across edge cameras by deploying a scalable DeepStream pipeline to run YOLO models via gRPC streaming of inference outputs to Google BigQuery for cross-run tracking

Publications

AMMPER-2: A spatially explicit agent-based model of microbial radiobiology with redox dye simulation
Daniel Palacios*, Pramesh Sharma*, Madeline Marous, Lauren Liddel, Diana Gentry, Sergio Santa Maria, Jessica A. Lee
https://doi.org/10.64898/2026.10.08.757403

AlphaFold reveals but sometimes distorts an organizational principle of protein folding 01/2026
Nature Communications
Madeleine F Clore, Joseph F Thole, Suchetan Dontha, Pramesh Sharma, Naomi Greenberg, Marie-Paule Strub, Mary Starich, Davin Jensen, Brian F Volkman, Matthew Coudron, Lauren L Porter
https://pubmed.ncbi.nlm.nih.gov/41509248/

EasyDIVER+: an advanced tool for analyzing high throughput sequencing data from in vitro evolution of nucleic acids or amino acids
04/2025
Journal of Molecular Evolution
Celia Blanco, Allison Tee, Pramesh Sharma, Matilda S. Newton, Kun-Hwa Lee, Samuel E. Erickson, Burckhard Seelig, Irene A. Chen
https://pmc.ncbi.nlm.nih.gov/articles/PMC12006205/

Internships

NASA 08/2023 - 12/2023
Computational Biology Intern
- Conducted radiobiology literature review on the linear quadratic model and integrated it into an agent based simulation aligning radiation dose-response with experimental data and biological phenomena of deep space effects on yeast cells
- Redefined cell health model using a systems framework to simulate and enhance cell behavior predictions, establishing probability distributions for apoptosis, DNA damage, and mitotic catastrophe to gamma and deep space radiation conditions
- Enhanced model accuracy by designing a metabolic viability model using Michaelis-Menten kinetics leveraging an evolutionary optimization strategy to finetune hyperparameters across random trials

NASA 08/2022 - 12/2022
Space Systems Engineering Intern
- Streamlined cross collaboration by standardizing project modeling requirements for the Lunar Pilot Excavator through a systems engineering modeling management plan in MagicDraw enabling 8+ engineers to work from a single source of truth
- Replaced paper-based workflows across 3+ engineering divisions through automated reports built with the velocity templage language aligning digital engineering review, procurement, and subsystem design processes
- Communicated project modeling strategy to chief and systems engineering stakeholders by presenting systems engineering template and concept of operations in final design reviews with a live demo demonstrating digital resusability

Genentech 06/2022 - 09/2022
Computational Proteomics Intern
- Increased efficiency of antibody analysis for wet lab scientists by developing an automated internal tool using a Python REST framework that converts mass spectrometry data into heat maps for visualization
- Streamlined data processing for large datasets by creating a command line interface using the optparse R package allowing users to handle up to 3,000 samples of protein-peptide entries per antibody analysis runs
- Accelerated pipeline deployment by containerizing tool dependencies with Docker, reducing discrepancies between dev and production environments

Genentech 01/2022 - 06/2022
Pharma Technical Operations Intern
- Increased efficiency of drug campaign creation by optimizing manufacturing on-call service and engineering assignments through a digital scheduling dashboard using JavaScript and PagerDuty API
- Automated user access to drug campaign incidents with JavaScript and Google Sheets API, resulting in manufacturing cost-savings of 5-minutes per incident and reliable lookup across process engineering functions
- Standardized Genentech's PagerDuty response procedures by documenting a comprehensive 10-page Rapid Response Playbook ensuring SOP compliance and producing a step by step video demo for tool best practices